Future work: testing the open boundaries I

The staged research program follows directly from the boundaries in sec. 8.13. It separates public-library reproducibility, server theory, portable source-protocol work, task generalization, and deployment validation so that success in one lane cannot silently certify another.

The order is deliberate. Simulation-study guidance recommends declaring the estimand, data-generating mechanism, and Monte Carlo precision before treating replication as evidence (Morris et al. 2019), while Monte Carlo error should be reported separately from an interval or a hypothesis test (Koehler et al. 2009). For nested agents, trials, and seeds, the resampling unit must respect the dependence structure (Loy and Korobova 2021). Accordingly, the phase plan records a primary unit and a falsifier for each extension; a larger sample or more elaborate diagram is not itself a stronger claim.

Future work: testing the open boundaries I

The implementation registry also separates smoke, pilot, and confirmatory profiles. Pilot worlds select budgets and calibration settings but never enter confirmatory intervals or headline values. Each completed run must bind its source bundle, configuration, dataset bytes, device, checkpoints, outputs, and completion status into a verifiable receipt. A negative or null scientific result remains a valid citable outcome when those implementation and provenance gates pass; it blocks the intended positive claim, not the software release.

Future work: testing the open boundaries I

A separate Friston protocol lane will resolve a machine-readable parity matrix before reconstructing source experiments in Python. Until every required source parameter, routine, unit, and estimand is recovered, that lane will be described as a paper-constrained reconstruction rather than an exact replication.

The asymmetry between the robustness axes (sec. 3.3) is the most consequential open problem. The client-side β/rcce update carries a derived, loss-specific bounded-influence result under the matching assumptions; the sharp server-side `robust_aggregate` carries only its recovery limit (eq. 7); and the new variational aggregator (sec. 5.8.2, sec. 11) supplies a *rigorous* server-side rule — exact block updates descending the stated free energy eq. 8 through non-increasing block updates, with a proven raw effective-weight bound and the same recovery corner. What it costs is conservatism: it is the maximum-entropy consensus, not the sharp accuracy-maximizer. The remaining open problem is therefore sharper than before — to write down a generalized variational objective in the FedGVI family (Mildner et al. 2025), informed by recent closed-form GVI characterizations

(Nguyen and Westerhout 2026), logarithmic-pool weighting theory (Carvalho et al. 2023), and robust divergence-weighted federated aggregation (Li et al. 2022), whose closed-form minimizer is competitive with the empirical reweighting across declared contamination regimes. That would combine axis 3's effective-weight bound with axis 2's empirical sharpness in one server rule. The recovery corner and the variational objective together supply two boundary conditions any such unification must satisfy.

Any empirical choice of `robustness` or `entropy_weight` will be made on separate calibration worlds using a proper log score, then frozen before confirmatory evaluation. This guards against selecting an apparent leader with evaluation truth and preserves null or reversed confirmatory outcomes. The comparison family will include logarithmic and linear pools, the current heuristic, the variational family, and a centered-log-ratio geometric-median control; none inherits a parameter-space robust-federated-learning guarantee merely by operating on beliefs.

Promote the baseline to original FedGVI scale I

The bounded-influence result is anchored locally by the small NumPy logistic-regression baseline (fig. 16). Promoting it to the GPU-scale Bayesian-neural-network experiments of the source paper (Mildner et al. 2025) — the experiments deferred here (sec. 8.13) — would test whether the per-client robustness curve holds at the model capacity and contamination regimes where federated learning (McMahan et al. 2017) actually operates, and would connect the discrete-POMDP result to the partitioned-VI line (Ashman et al. 2022; Bui et al. 2018). The planned comparison would also require posterior-parameterization parity with Bayesian neural-network work, rather than treating the current deterministic point-mass MLP as a posterior (Mildner et al. 2025).

Promote the baseline to original FedGVI scale I

The portable lane preserves the source protocol's site factors, client cavity, and factor-replacement update in natural coordinates. It distinguishes a locally budgeted CPU/MPS profile from an exact source-scale CUDA profile that remains external until suitable hardware is available. FashionMNIST anchors protocol parity, while MNIST and KMNIST test portability. A separate source-bound tabular pack will report proper-score effects per licensed dataset, with training-only preprocessing and byte-, split-, and license-level provenance; nested seeds will not be treated as independent datasets.

The governing caveat here is that added depth must be shown to *earn* generalization, not merely to execute: on the current sentinel task the deeper stacks match rather than beat the flat baseline on location (sec. 8.15). The extension therefore has two distinct fronts — carrying the recovery contract to deeper stacks, and finding a task family in which depth actually pays.

The 2-level hierarchical POMDP (sec. 13.9) couples location inference to a single global context. The generic N-level architecture (:func:`fedference.pomdp.build_nlevel_world`) has already been exercised with a 3-level stack (sec. 13.11, sec. 13.12): a meta-context variable (L3) gates the context prior (L2) which in turn gates the location prior (L1). The empirical-prior top-down messages (eq. 32, eq. 33) remain valid variational steps at every depth, and the log-linear-pool federation at each level is bit-identical to the in-process result (Proposition 12). The natural next question is whether the limit-as-proof contract of sec. 8.13 survives still deeper hierarchies: does the recovery corner (context prior $\rightarrow$ uniform) remain checkable to machine precision when the L2 $\rightarrow$ L1 message is itself a function of an L3 belief coupled to an L4 belief, and can message-passing engines such as RxInfer (Bagaev et al. 2023) carry the generic alternating-minimization at scale? Structure learning already answers the dual question — how *deep* the model should be — for the top level: hierarchical Bayesian model reduction (sec. 14.1) prunes a non-gating meta-context and keeps an informative one, so the depth is decided by the evidence rather than assumed. Extending that per-level reduction to a full breadth-and-depth search over the generic N-level stack is the natural continuation.

Extend hierarchical federation beyond the current stack I

The next task family is deliberately controlled rather than merely deeper: partially observable Four Rooms and Key-Door will compare flat, oracle, learned, shuffled, and non-gating hierarchies at matched horizons and compute. Task is the higher-level replication unit. A gain in only one task will remain task-specific, and the larger campaign will not begin until the hybrid representation recovery gates pass.

Promoting federation from the current queue-backed, single-machine process and loopback-socket helpers to cross-host workers would retire the remaining deployment caveat of sec. 8.13 while preserving the bit-identical consensus property proved in Proposition 12. The `federation/` package and federation tests already establish the API contract: a server collects serialized beliefs from 5 worker channels, fuses them with `robust_aggregate` at robustness $c = 1.5$, and broadcasts the consensus back over response channels, with bit-identity verified at True. The loopback socket path adds optional pre-shared-key frame integrity and file-backed digest-verified replay validation. The next systems step is an explicitly local Docker multi-node emulator with mTLS by default, HMAC compatibility, checkpoint/restart, and reproducible message-fault controls. It is not physical multi-host evidence. That later claim requires receipts from distinct hosts, deployment-grade key management, timeout policy, and long-running orchestration across process restarts, but it does not require changing the mathematics. Secure aggregation, differential privacy, and Byzantine tolerance are separate future threat models; transport integrity alone would not establish any of them (Blanchard et al. 2017; Pillutla et al. 2022).

Move beyond categorical state spaces I

This work is discrete-categorical, matching the community's worked POMDP (Da Costa et al. 2020). A first step is already in place: the closed-form Gaussian KL and Rényi divergences of sec. 12 show the divergence family — and its $\alpha \to 1$ recovery — carries over verbatim to Gaussian beliefs, scoped out of the categorical experiments. Continuous-state Gaussian generative models would test whether the limit-as-proof contract survives the move off categorical state spaces — whether the recovery corner remains checkable to machine precision when the belief simplex is replaced by a Gaussian belief, and whether message-passing engines such as RxInfer (Bagaev et al. 2023) can carry the robust update at scale. Extending the structure-learning study (Smith et al. 2022; Friston and Penny 2011) into continuous models would, in parallel, test whether robust belief fusion and robust *structure* fusion compose.

A minimal executable fixture now gates a discrete dynamics context over continuous position and velocity, Gaussian observations, and bounded actions. It is a representation and recovery surface, not confirmatory task evidence. The full study must add discrete-only, continuous-only, and oracle-context controls, singular-covariance and outlier checks, and held-out posterior-predictive scoring before supporting a hybrid-task claim.